

CIMPA ECCO 2026 Worksheets and Solutions

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1 Worksheet 1

1. Find all permutations $\pi \in S_4$ for which $\pi(f) = f$, where $f(x_1, x_2, x_3, x_4) = x_1x_2^2x_4^4 + x_2^2x_3^4x_4 + x_1^4x_2^2x_3$.
2. Find a polynomial f in x_1, x_2, x_3 which has $\pi(f) = \text{sgn}(\pi)f$ for all $\pi \in S_3$, and which has $x_1^7x_3^2$ as one of its terms. Recall that $\text{sgn}(\pi)$ is the **sign** of the permutation that is 1, if π is the product of an even number of transpositions and -1 if π is the product of an odd number of permutations.
3. Compute the four monomial symmetric polynomials $m_{3,2}(x_1, x_2)$, $m_{3,2}(x_1, x_2, x_3)$, $m_{3,3,1,1}(x_1, x_2, x_3)$, and $m_{3,3,1,1}(x_1, x_2, x_3, x_4)$.
4. Compute the product of the formal power series $f(x) = \sum_{i=0}^{\infty} x^i$ and $g(x) = \sum_{j=0}^{\infty} jx^j$.
5. Compute the product of the formal power series $f = \sum_{i=1}^{\infty} x_i x_{i+1}$ and $g = \sum_{j=1}^{\infty} x_j$.
6. Compute $m_{2,1}$. Is $m_2 m_1 = m_{2,1}$?
7. Using symmetry, write $m_3 m_2 m_1$ as a linear combination of $\{m_\lambda : \lambda \vdash 6\}$.
8. Without multiplying everything out, find the coefficient of $m_{2,1^{a+b}}$ in the product $m_{1^{a+1}} m_{1^{b+1}}$.
9. Write the following as a sum of monomial symmetric functions:
 - (a) $\prod_{i=1}^{\infty} (1 + x_i)$.
 - (b) $\prod_{i=1}^{\infty} (1 - x_i)$
10. Use the geometric series to write a very simple infinite product using the variables x_i 's that is equal to the sum

$$\sum_{\lambda} m_{\lambda},$$

where here the sum is over all partitions of all positive integers.

Español by Ignacio Rojas

1. Encuentre todas las permutaciones $\pi \in S_4$ para las cuales $\pi(f) = f$, donde $f(x_1, x_2, x_3, x_4) = x_1x_2^2x_4^4 + x_2^2x_3^4x_4 + x_1^4x_2^2x_3$.
2. Encuentre un polinomio f en x_1, x_2, x_3 que satisfaga $\pi(f) = \text{sgn}(\pi)f$ para toda $\pi \in S_3$, y que tenga $x_1^7x_3^2$ como uno de sus términos. Recuerde que $\text{sgn}(\pi)$ es el **signo** de la permutación, que es 1 si π es el producto de un número par de transposiciones y -1 si π es el producto de un número impar de permutaciones.

3. Calcule los cuatro polinomios simétricos monomiales $m_{3,2}(x_1, x_2)$, $m_{3,2}(x_1, x_2, x_3)$, $m_{3,3,1,1}(x_1, x_2, x_3)$ y $m_{3,3,1,1}(x_1, x_2, x_3, x_4)$.
4. Calcule el producto de las series de potencias formales $f(x) = \sum_{i=0}^{\infty} x^i$ y $g(x) = \sum_{j=0}^{\infty} jx^j$.
5. Calcule el producto de las series de potencias formales $f = \sum_{i=1}^{\infty} x_i x_{i+1}$ y $g = \sum_{j=1}^{\infty} x_j$.
6. Calcule $m_{2,1}$. ¿Vale que $m_2 m_1 = m_{2,1}$?
7. Usando la simetría, escriba $m_3 m_2 m_1$ como una combinación lineal de $\{m_\lambda : \lambda \vdash 6\}$.
8. Sin multiplicar todo, encuentre el coeficiente de $m_{2,1^{a+b}}$ en el producto $m_{1^{a+1}} m_{1^{b+1}}$.
9. Reescriba lo siguiente como una suma de funciones simétricas monomiales:
 - (a) $\prod_{i=1}^{\infty} (1 + x_i)$.
 - (b) $\prod_{i=1}^{\infty} (1 - x_i)$
10. Utilizando la serie geométrica, escriba un producto infinito muy sencillo usando las variables x_i que sea igual a la suma

$$\sum_{\lambda} m_{\lambda},$$

donde aquí la suma es sobre todas las particiones de todos los enteros positivos.

2 Worksheet 2

1. Compute $M_{\lambda,\mu}(e, m)$ for $\lambda = (2^2, 1^4)$ and $\mu = (3^2, 1^2)$.
2. Find and prove a formula for $M_{\lambda, 1^n}(e, m)$, for any $\lambda \vdash n$.
3. Find and prove a formula for $M_{\lambda, n}(e, m)$, for any $\lambda \vdash n$.
4. Find and prove a formula for $M_{\lambda, (n-1, 1)}(e, m)$, for any $\lambda \vdash n$.
5. Prove or disprove: If λ and μ are partitions with $|\lambda| = |\mu|$ and $\lambda >_{\text{lex}} \mu$, then $\mu' >_{\text{lex}} \lambda'$.
6. Write the formal power series

$$\prod_{i=1}^{\infty} (1 + x_i + x_i^2)$$

in terms of the elementary symmetric functions.

7. There is another combinatorial interpretation for $M_{\lambda,\mu}(e, m)$, if $\lambda, \mu \vdash k$, $M_{\lambda,\mu}(e, m)$ is equal to the number of $k \times k$ matrices with entries 0 or 1 only, such that the sum of the i -th row is μ_i and the sum of column j is λ_j for all i and j .
 - (a) Use this combinatorial interpretation to find $M_{\lambda,\mu}(e, m)$ for $\lambda = (4, 2, 1)$ and $\mu = (3, 1^4)$.
 - (b) Use this interpretation to show that $M_{\lambda,\mu}(e, m) = M_{\mu,\lambda}(e, m)$.
 - (c) Describe a bijection from the tableaux interpretation to the $\{0, 1\}$ -matrix interpretation.
8. Compute $M_{\lambda,\mu}(h, m)$ for $\lambda = (2^2, 1^4)$ and $\mu = (3^2, 1^2)$.
9. Find and prove a formula for $M_{\lambda, 1^n}(h, m)$, for any $\lambda \vdash n$.
10. Find and prove a formula for $M_{\lambda, n}(h, m)$, for any $\lambda \vdash n$.
11. Find and prove a formula for $M_{\lambda, (n-1, 1)}(h, m)$, for any $\lambda \vdash n$.
12. Show that for all $k \geq 1$ and all partitions $\lambda, \mu \vdash k$, the quantity $M_{\lambda,\mu}(p, m)$ is the number of fillings of the Young diagram of λ with positive integers for which the entries in each row are constant, and each integer i appears μ_i times.
13. Compute $M_{\lambda,\mu}(p, m)$ for $\lambda = (2^2, 1^4)$ and $\mu = (3^2, 1^2)$.
14. Find and prove a formula for $M_{\lambda, 1^n}(p, m)$, for any $\lambda \vdash n$.
15. Find and prove a formula for $M_{\lambda, n}(p, m)$, for any $\lambda \vdash n$.
16. Find and prove a formula for $M_{\lambda, (n-1, 1)}(p, m)$, for any $\lambda \vdash n$.

Español

1. Calcule $M_{\lambda,\mu}(e, m)$ para $\lambda = (2^2, 1^4)$ y $\mu = (3^2, 1^2)$.
2. Encuentre y pruebe una fórmula para $M_{\lambda,1^n}(e, m)$, para cualquier $\lambda \vdash n$.
3. Encuentre y pruebe una fórmula para $M_{\lambda,n}(e, m)$, para cualquier $\lambda \vdash n$.
4. Encuentre y pruebe una fórmula para $M_{\lambda,(n-1,1)}(e, m)$, para cualquier $\lambda \vdash n$.
5. Pruebe o dé un contraejemplo: Si λ y μ son particiones con $|\lambda| = |\mu|$ y $\lambda >_{\text{lex}} \mu$, entonces $\mu' >_{\text{lex}} \lambda'$.
6. Escriba la serie de potencias formal

$$\prod_{i=1}^{\infty} (1 + x_i + x_i^2)$$

en términos de las funciones simétricas elementales.

7. Existe otra interpretación combinatoria para $M_{\lambda,\mu}(e, m)$: si $\lambda, \mu \vdash k$, $M_{\lambda,\mu}(e, m)$ es igual al número de matrices $k \times k$ cuyas entradas son únicamente 0 o 1, tales que la suma de la i -ésima fila es μ_i y la suma de la columna j es λ_j para todos i y j .
 - (a) Use esta interpretación combinatoria para encontrar $M_{\lambda,\mu}(e, m)$ para $\lambda = (4, 2, 1)$ y $\mu = (3, 1^4)$.
 - (b) Use esta interpretación para demostrar que $M_{\lambda,\mu}(e, m) = M_{\mu,\lambda}(e, m)$.
 - (c) Describa una biyección entre la interpretación como tablas y la interpretación como matrices de $\{0, 1\}$.
8. Calcule $M_{\lambda,\mu}(h, m)$ para $\lambda = (2^2, 1^4)$ y $\mu = (3^2, 1^2)$.
9. Encuentre y pruebe una fórmula para $M_{\lambda,1^n}(h, m)$, para cualquier $\lambda \vdash n$.
10. Encuentre y pruebe una fórmula para $M_{\lambda,n}(h, m)$, para cualquier $\lambda \vdash n$.
11. Encuentre y pruebe una fórmula para $M_{\lambda,(n-1,1)}(h, m)$, para cualquier $\lambda \vdash n$.
12. Demuestre que para todo $k \geq 1$ y todas las particiones $\lambda, \mu \vdash k$, $M_{\lambda,\mu}(p, m)$ es el número de rellenos de Tablas de Young de λ **UNSURE** con enteros positivos, en los cuales las entradas de cada fila son constantes y cada entero i aparece μ_i veces.
13. Calcule $M_{\lambda,\mu}(p, m)$ para $\lambda = (2^2, 1^4)$ y $\mu = (3^2, 1^2)$.
14. Encuentre y pruebe una fórmula para $M_{\lambda,1^n}(p, m)$, para cualquier $\lambda \vdash n$.
15. Encuentre y pruebe una fórmula para $M_{\lambda,n}(p, m)$, para cualquier $\lambda \vdash n$.
16. Encuentre y pruebe una fórmula para $M_{\lambda,(n-1,1)}(p, m)$, para cualquier $\lambda \vdash n$.

3 Worksheet 3

1. Write the Schur function $s_{2,2}$ as a linear combination in the monomial, homogeneous, elementary and power symmetric functions.
2. Prove the following for $\lambda, \mu \vdash k$,
 - (a) If $\mu >_{\text{lex}} \lambda$, then $K_{\lambda, \mu} = 0$
 - (b) $K_{\lambda, \lambda} = 1$.
3. Use Exercise 2 to prove that $\{s_\lambda : \lambda \vdash k\}$ is a basis for Λ^k .
4. How many semistandard tableaux (of any shape) have content k, k , i.e., this means k 1's and k 2's.
5. Prove that ω is an involution using the identity that for $n \geq 1$,

$$\sum_{i=0}^n (-1)^i e_i h_{n-i} = 0.$$

6. Write $h_{3,2}$ in the Schur basis.
7. Write $e_{3,2}$ in the Schur basis.
8. Write p_3 in the Schur basis.
9. Prove or disprove: for all $\lambda, \mu \vdash k$, we have $K_{\lambda, \mu} = K_{\mu, \lambda}$.
10. For each partition of $\lambda \vdash 3$, write h_λ as a linear combination of Schur functions. Make a conjecture about the coefficients you obtain.
11. Use the Jacobi-Trudi identity and induction to prove that for all n and k with $n \geq k \geq 0$, we have

$$s_{a, 1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}$$

Español

1. Escriba la función de Schur $s_{2,2}$ como una combinación lineal de las funciones simétricas monomiales, homogéneas, elementales y de potencias.
2. Demuestre lo siguiente para $\lambda, \mu \vdash k$,
 - (a) Si $\mu >_{\text{lex}} \lambda$, entonces $K_{\lambda, \mu} = 0$.
 - (b) $K_{\lambda, \lambda} = 1$.
3. Use el Ejercicio 2 para demostrar que $\{s_\lambda : \lambda \vdash k\}$ es una base para Λ^k .

4. ¿Cuántas tablas semiestándar (de cualquier forma) tienen contenido k, k , es decir, tienen k 1's y k 2's?
5. Demuestre que ω es una involución usando la identidad que para $n \geq 1$,

$$\sum_{i=0}^n (-1)^i e_i h_{n-i} = 0.$$

6. Escriba $h_{3,2}$ en la base de Schur.
7. Escriba $e_{3,2}$ en la base de Schur.
8. Escriba p_3 en la base de Schur.
9. Pruebe o dé un contraejemplo: para $\lambda, \mu \vdash k$, tenemos $K_{\lambda, \mu} = K_{\mu, \lambda}$.
10. Para cada partición de $\lambda \vdash 3$, escriba h_λ como una combinación lineal de funciones de Schur. Formule una conjetura sobre los coeficientes que obtiene.
11. Con la identidad de Jacobi-Trudi e inducción, pruebe que para n y k con $n \geq k \geq 0$, tenemos

$$s_{a,1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}$$

4 Worksheet 4

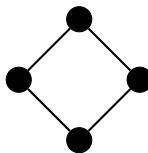
1. Explain why the chromatic symmetric function is symmetric.
2. Let G be a graph and e an edge in G . In graph theory the chromatic polynomial is computed using the deletion-contraction recurrence:

$$\chi_G(n) = \chi_{G-e}(n) - \chi_{G/e}(n),$$

where $G - e$ is the graph with the edge e removed and with the same vertices as G , and G/e is the graph with the edge e contracted to single vertex. If multiple edges result, you replace them with a single edge. Use this recurrence to compute the chromatic polynomial of the cycle graph with four vertices (the image is in Problem #3).

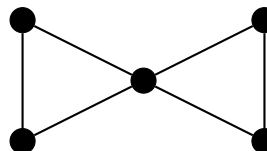
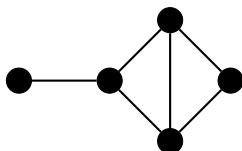
Recall that the chromatic polynomial of a tree, T , with k vertices is $\chi_T(n) = n(n-1)^{k-1}$

3. Take the cycle graph with four vertices and choose one edge.

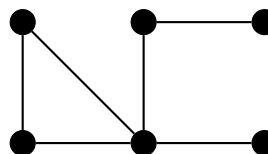
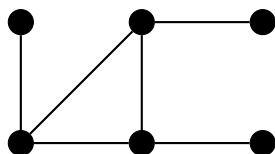


- (a) What graph do you get when you delete this edge? Compute the chromatic symmetric function for the graph that you get in the power sum basis.
 - (b) What graph do you get when you contract this edge? Compute the chromatic symmetric function for the graph that you get in the power sum basis.
 - (c) Using this example, explain why the chromatic symmetric function does not satisfy the deletion-contraction recurrence.
4. A tree with k vertices can be defined as a connected graph with $k - 1$ edges.
 - (a) In our lecture we said that all trees have chromatic polynomial equal to $n(n-1)^{k-1}$. Use the binomial theorem, $(a+b)^m = \sum_{i=0}^m \binom{m}{i} a^i b^{m-i}$, to expand this polynomial.
 - (b) What is the coefficient of n^{k-1} ?
 - (c) Use induction and the deletion-contraction recurrence to show that the coefficient of n^{k-1} in $\chi_G(n)$ is equal to the number of edges in G .
 - (d) Suppose that G is a connected graph with k vertices and $\chi_G(n) = n(n-1)^{k-1}$, explain why G must be a tree.

5. Compute the chromatic symmetric function of the following graphs in the monomial basis.



6. Given X_G for some graph G in the power basis, how can you tell whether G is connected or not from the coefficients?
7. Show that if G and H satisfy $X_G = X_H$, then G and H must have the same number of edges.
8. Compute the chromatic symmetric function of the complete graph for $n \leq 5$ vertices in the elementary basis. Conjecture a formula for the chromatic symmetric function of the complete graph.
9. Let G be a graph with a leaf edge e ,
- Describe the graph that you get when you delete e .
 - Describe the graph that you get when you do a dot-contraction to e .
 - How are the graphs that you got in (a) and in (b) related?
 - What graph do you get when you leaf-contrast the edge e ?
 - How does this justify that we only apply deletion-near-contraction to internal edges?
10. Use the deletion-near-contraction relation to find the chromatic symmetric function of all trees with 5 vertices in the star basis.
11. Compute the chromatic symmetric function of the following two graphs in the star basis.



Español

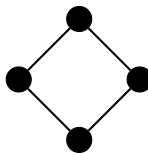
1. Explique por qué la función simétrica cromática es una función simétrica.
2. Sea G un grafo y e una arista en G . En teoría de grafos, el polinomio cromático se calcula usando la relación de eliminación-contracción:

$$\chi_G(n) = \chi_{G-e}(n) - \chi_{G/e}(n),$$

donde $G - e$ es el grafo con la arista e eliminada y con los mismos vértices que G , y G/e es el grafo que se obtiene al contraer la arista e a un solo vértice. Si resultan múltiples aristas, reemplácelas por una sola arista. Usando esta relación, calcule el polinomio cromático del grafo cíclico de cuatro vértices (la imagen del Problema #3).

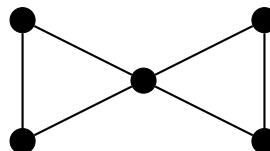
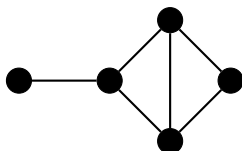
Recuerde que el polinomio cromático de un árbol T con k vértices es $\chi_T(n) = n(n-1)^{k-1}$.

3. Tome el grafo cíclico con cuatro vértices y elija una arista.

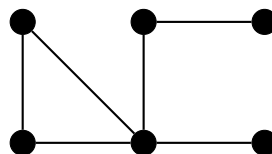
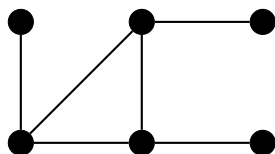


- (a) ¿Qué grafo obtiene al eliminar esta arista? Calcule la función simétrica cromática del grafo obtenido en la base de sumas de potencias.
 - (b) ¿Qué grafo obtiene al contraer esta arista? Calcule la función simétrica cromática del grafo obtenido en la base de sumas de potencias.
 - (c) Usando este ejemplo, explique por qué la función simétrica cromática no satisface la relación de eliminación-contracción.
4. Un árbol con k vértices puede definirse como un grafo conexo con $k - 1$ aristas.
 - (a) En clase dijimos que todos los árboles tienen polinomio cromático igual a $n(n-1)^{k-1}$. Con el teorema del binomio, $(a+b)^m = \sum_{i=0}^m \binom{m}{i} a^i b^{m-i}$, expanda este polinomio.
 - (b) ¿Cuál es el coeficiente de n^{k-1} ?
 - (c) Use inducción y la relación de eliminación-contracción para demostrar que el coeficiente de n^{k-1} en $\chi_G(n)$ es igual al número de aristas en G .
 - (d) Suponga que G es un grafo conexo con k vértices y $\chi_G(n) = n(n-1)^{k-1}$; explique por qué G debe ser un árbol.

5. Calcule la función simétrica cromática de los siguientes grafos en la base monomial.



6. Dado X_G para algún grafo G en la base de potencias, ¿cómo puede determinar a partir de los coeficientes si G es conexo o no?
7. Demuestre que si G y H satisfacen $X_G = X_H$, entonces G y H deben tener el mismo número de aristas.
8. Calcule la función simétrica cromática del grafo completo para $n \leq 5$ vértices en la base elemental. Conjeture una fórmula para la función simétrica cromática del grafo completo.
9. Sea G un grafo con una hoja e ,
- Describa el grafo que obtiene al eliminar e .
 - Describa el grafo que obtiene al hacer una contracción por punto de e .
 - ¿Cómo se relacionan los grafos que obtuvo en (a) y (b)?
 - ¿Qué grafo obtiene al contraer-por-hoja (*leaf-contract*) la arista e ?
 - ¿Cómo justifica esto que solo apliquemos la eliminación-cerca-de-contracción (*deletion-near-contraction*) a aristas internas?
10. Use la relación de eliminación-cerca-de-contracción (*deletion-near-contraction*) para encontrar la función simétrica cromática de todos los árboles con 5 vértices en la base de estrellas (*star basis*).
11. Calcule la función simétrica cromática de los siguientes dos grafos en la base de estrellas.



5 SageMath Worksheet

1. For every partition n , $n \leq 6$, compute the change of basis matrix from the m -basis to the s -basis.
 - (a) What ordering is Sage using?
 - (b) Is the matrix triangular?
 - (c) What is the determinant?
2. Choose several partitions λ and compute $s[\lambda].\text{expand}(4)$ for each. Make sure your examples include partitions with more and fewer than 4 parts.
 - (a) What does this function do?
 - (b) In each of your examples, use these expressions to write $s[\lambda].\text{expand}(4)$ as a sum of monomial symmetric functions.
 - (c) Look for patterns in the coefficients. What do you notice?
3. For $n \leq 7$ vertices, find the chromatic symmetric function of the following class of graphs: paths, cycles, and complete graphs.
 - (a) For each class, expand in the monomial, elementary, power, and Schur bases.
 - (b) Which basis has the simplest expression for each class of graphs?
 - (c) Does the chromatic symmetric function have all positive coefficients in any of the bases?
4. Compute the chromatic symmetric function for the star graph with n vertices, St_n , for $1 \leq n \leq 7$. Check the proposition about the expression $\mathbf{X}_{St_n}$ in the power symmetric function.
5. Generate all trees with $n = 7$ vertices and compute their chromatic symmetric functions. Are there two trees with the same function? Repeat with $n = 8$.
6. Expand the chromatic symmetric function of each tree on $n = 8$ vertices and write it in the elementary basis. Does the coefficient of $e_{4,2,1,1}$ distinguish these trees?
7. For every tree with 8 vertices, compute the following:
 - (a) degree sequence.
 - (b) diameter.
 - (c) number of leaves.
 - (d) the coefficient of $e_{n-1,1}$ and $p_{n-1,1}$.
 - (e) Are there any correlations between the coefficients and items (a) - (c)?

Mini-Projects

A. **Question:** Which basis best encode trees?

Objective: Investigate how the chromatic symmetric function behaves in each of the bases.

- (a) Compute the chromatic symmetric function for each tree with $n = 8$ vertices.
- (b) For each tree, write its chromatic symmetric function in the monomial, elementary, power, and Schur basis.
- (c) Store the number of nonzero coefficients, largest coefficient, smallest coefficient.
- (d) Create a table such that list every tree and number of terms in each of the basis.
- (e) Which basis gives the sparsest expansion?
- (f) Is there a basis with all positive coefficients?
- (g) Is there anything that you can tell about the tree from the coefficients?

B. **Question:** What is the smallest and largest partitions in lexicographic order that appears with nonzero coefficient?

Objective: See if these partitions or their coefficients give information about the trees.

- (a) Compute the chromatic symmetric function for each tree with $n = 8$ vertices.
- (b) Expand the chromatic symmetric functions in the monomial basis.
- (c) For each tree write the smallest and largest partition in lexicographic largest and smallest partition.
- (d) Does it encode any information about the tree?
- (e) How does the coefficient change when you add a leaf to your tree?
- (f) Repeat the whole exercise for the other bases of symmetric functions.

Español

1. Para cada partición de n , con $n \leq 6$, calcule la matriz de cambio de base de la base m a la base s .
 - (a) ¿Qué orden está usando Sage?
 - (b) La matriz que resulta, ¿es triangular?
 - (c) ¿Cuál es su determinante?
2. Elija varias particiones λ y calcule `s[λ].expand(4)` para cada una. Asegúrese de que sus ejemplos incluyan particiones con más y menos de 4 partes.
 - (a) ¿Qué hace esta función?
 - (b) En cada uno de sus ejemplos, use estas expresiones para escribir `s[λ].expand(4)` como una suma de funciones simétricas monomiales.
 - (c) Busque patrones en los coeficientes. ¿Qué observa?
3. Para $n \leq 7$ vértices, encuentre la función simétrica cromática de las siguientes clases de grafos: caminos, ciclos y grafos completos.
 - (a) Para cada clase, expanda en las bases monomial, elemental, de potencias y de Schur.
 - (b) ¿Qué base tiene la expresión más sencilla para cada clase de grafos?
 - (c) ¿Tiene la función simétrica cromática todos sus coeficientes positivos en alguna de las bases?
4. Calcule la función simétrica cromática del grafo estrella con n vértices, St_n , para $1 \leq n \leq 7$. Compruebe la proposición sobre la expresión $\mathbf{X}_{St_n}$ en la función simétrica de potencias.
5. Genere todos los árboles con $n = 7$ vértices y calcule sus funciones simétricas cromáticas. Existen dos árboles con la misma función? Repita con $n = 8$.
6. Expanda la función simétrica cromática de cada árbol con $n = 8$ vértices y escríbala en la base elemental. ¿Distingue el coeficiente de $e_{4,2,1,1}$ a estos árboles?
7. Para cada árbol con 8 vértices, calcule lo siguiente:
 - (a) la sucesión de grados.
 - (b) el diámetro.
 - (c) el número de hojas.
 - (d) el coeficiente de $e_{n-1,1}$ y $p_{n-1,1}$.
 - (e) ¿Hay alguna correlación entre los coeficientes y los elementos (a)–(c)?

Mini-Proyectos

A. **Pregunta:** ¿Qué base codifica mejor los árboles?

Objetivo: Investigar cómo se comporta la función simétrica cromática en cada una de las bases.

- (a) Calcule la función simétrica cromática de cada árbol con $n = 8$ vértices.
- (b) Para cada árbol, escriba su función simétrica cromática en las bases monomial, elemental, de potencias y de Schur.
- (c) Guarde el número de coeficientes no nulos, el coeficiente más grande y el coeficiente más pequeño.
- (d) Cree una tabla que enumere cada árbol y el número de términos en cada una de las bases.
- (e) ¿Qué base produce la expansión más esparza (*sparsest*)?
- (f) ¿Existe una base con todos sus coeficientes positivos?
- (g) ¿Hay algo que se pueda decir sobre el árbol a partir de los coeficientes?

B. **Pregunta:** ¿Cuáles son las particiones más pequeña y más grande en orden lexicográfico que aparecen con coeficiente no nulo?

Objetivo: Ver si estas particiones o sus coeficientes proporcionan información sobre los árboles.

- (a) Calcule la función simétrica cromática de cada árbol con $n = 8$ vértices.
- (b) Expanda las funciones simétricas cromáticas en la base monomial.
- (c) Para cada árbol, escriba la partición más pequeña y la más grande en orden lexicográfico.
- (d) ¿Proporciona esto alguna información sobre el árbol?
- (e) ¿Cómo cambia el coeficiente cuando añade una hoja a su árbol?
- (f) Repita todo el ejercicio para las otras bases de funciones simétricas.

6 Solutions to Worksheet 1

1. **Solution.** We use the action in which $\pi(x_i) = x_{\pi(i)}$. Write the three monomials of f as

$$A = x_1 x_2^2 x_4^4, \quad B = x_2^2 x_3^4 x_4, \quad C = x_1^4 x_2^2 x_3.$$

The variable x_2 is the only variable that has exponent 2 in every monomial. Therefore, any permutation that fixes f must fix 2.

The cycle $(1\ 4\ 3)$ fixes 2 and sends

$$A \mapsto B, \quad B \mapsto C, \quad C \mapsto A.$$

Thus $(1\ 4\ 3)$ fixes the sum f . Its inverse $(1\ 3\ 4)$ also fixes f , as does the identity.

There are no other possibilities. Indeed, after fixing 2, the image of A must be one of A , B , or C . Because the exponents 1, 0, and 4 on x_1, x_3, x_4 are distinct, each choice determines the permutation completely, and the three choices give precisely the permutations above. Hence

$$\boxed{\{\pi \in S_4 : \pi(f) = f\} = \{\text{id}, (1\ 4\ 3), (1\ 3\ 4)\}.$$

2. **Solution.** Start with the monomial $x_1^7 x_3^2$ and take its alternating sum over S_3 :

$$f = \sum_{\pi \in S_3} \text{sgn}(\pi) \pi(x_1^7 x_3^2).$$

Since the exponents 7, 2, and 0 are distinct, the six monomials in this sum are distinct. Expanding gives

$$\begin{aligned} f &= x_1^7 x_3^2 - x_2^7 x_3^2 - x_1^2 x_3^7 - x_1^7 x_2^2 \\ &\quad + x_1^2 x_2^7 + x_2^2 x_3^7. \end{aligned}$$

In particular, $x_1^7 x_3^2$ occurs with coefficient 1.

To verify the required property, let $\sigma \in S_3$. Reindexing the sum with $\tau = \sigma\pi$ gives

$$\begin{aligned} \sigma(f) &= \sum_{\pi \in S_3} \text{sgn}(\pi) (\sigma\pi)(x_1^7 x_3^2) \\ &= \text{sgn}(\sigma) \sum_{\tau \in S_3} \text{sgn}(\tau) \tau(x_1^7 x_3^2) = \text{sgn}(\sigma) f. \end{aligned}$$

Alternative solution. The same polynomial can be written as the determinant

$$f = \begin{vmatrix} x_1^7 & x_2^7 & x_3^7 \\ 1 & 1 & 1 \\ x_1^2 & x_2^2 & x_3^2 \end{vmatrix}.$$

Permuting the variables permutes the columns of the determinant, so the determinant is multiplied by the sign of the permutation. Expanding the determinant also shows that the coefficient of $x_1^7 x_3^2$ is 1.

3. **Solution.** A monomial symmetric polynomial is the sum of all distinct monomials obtained by permuting the indicated exponents among the available variables. Therefore,

$$m_{3,2}(x_1, x_2) = x_1^3 x_2^2 + x_1^2 x_2^3,$$

and

$$\begin{aligned} m_{3,2}(x_1, x_2, x_3) &= x_1^3 x_2^2 + x_1^3 x_3^2 + x_2^3 x_1^2 \\ &\quad + x_2^3 x_3^2 + x_3^3 x_1^2 + x_3^3 x_2^2. \end{aligned}$$

The partition $(3, 3, 1, 1)$ has four positive parts, so it cannot be placed in only three variables. Hence

$$m_{3,3,1,1}(x_1, x_2, x_3) = 0.$$

With four variables, choose the two variables that receive exponent 3; the other two receive exponent 1. Thus

$$\begin{aligned} m_{3,3,1,1}(x_1, x_2, x_3, x_4) &= x_1^3 x_2^3 x_3 x_4 + x_1^3 x_3^3 x_2 x_4 + x_1^3 x_4^3 x_2 x_3 \\ &\quad + x_2^3 x_3^3 x_1 x_4 + x_2^3 x_4^3 x_1 x_3 + x_3^3 x_4^3 x_1 x_2. \end{aligned}$$

4. **Solution.** The coefficient of x^n in the product is obtained by the Cauchy product:

$$\begin{aligned} f(x)g(x) &= \sum_{n=0}^{\infty} \left(\sum_{\substack{i+j=n \\ i,j \geq 0}} j \right) x^n \\ &= \sum_{n=0}^{\infty} \left(\sum_{j=0}^n j \right) x^n = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n. \end{aligned}$$

Equivalently, as formal power series,

$$f(x) = \frac{1}{1-x}, \quad g(x) = \frac{x}{(1-x)^2},$$

so

$$f(x)g(x) = \frac{x}{(1-x)^3} = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n.$$

5. **Solution.** Distributing the two sums gives

$$\begin{aligned} fg &= \left(\sum_{i=1}^{\infty} x_i x_{i+1} \right) \left(\sum_{j=1}^{\infty} x_j \right) \\ &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} x_i x_{i+1} x_j. \end{aligned}$$

Separating the cases $j = i$, $j = i + 1$, and $j \notin \{i, i + 1\}$ gives the more explicit form

$$fg = \sum_{i=1}^{\infty} (x_i^2 x_{i+1} + x_i x_{i+1}^2) + \sum_{i=1}^{\infty} \sum_{\substack{j \geq 1 \\ j \neq i, i+1}} x_i x_{i+1} x_j.$$

The last double sum should not be simplified by discarding repetitions. For example, $x_i x_{i+1} x_{i+2}$ occurs twice: once using the adjacent pair $(i, i + 1)$ and once using the adjacent pair $(i + 1, i + 2)$.

6. **Solution.** By definition,

$$m_{2,1} = \sum_{i \neq j} x_i^2 x_j = \sum_{i < j} (x_i^2 x_j + x_i x_j^2).$$

On the other hand,

$$\begin{aligned} m_2 m_1 &= \left(\sum_i x_i^2 \right) \left(\sum_j x_j \right) \\ &= \sum_i x_i^3 + \sum_{i \neq j} x_i^2 x_j = m_3 + m_{2,1}. \end{aligned}$$

Therefore,

$$m_2 m_1 \neq m_{2,1}; \quad m_2 m_1 = m_3 + m_{2,1}.$$

7. **Solution.** In a product $m_3 m_2 m_1$, choose one variable for the exponent 3, one for the exponent 2, and one for the exponent 1. We classify the result according to which choices are equal.

- If all three choices are the same, we obtain m_6 .
- If only the choices for exponents 3 and 2 are the same, we obtain $m_{5,1}$.
- If only the choices for exponents 3 and 1 are the same, we obtain $m_{4,2}$.
- If only the choices for exponents 2 and 1 are the same, we obtain $m_{3,3}$. Each monomial of this type occurs twice, because either of its two variables can come from the m_3 factor.
- If all three choices are distinct, we obtain $m_{3,2,1}$.

Consequently,

$$m_3 m_2 m_1 = m_6 + m_{5,1} + m_{4,2} + 2m_{3,3} + m_{3,2,1}.$$

8. **Solution.** Recall that

$$m_{1^r} = \sum_{\substack{A \subseteq \{1,2,\dots\} \\ |A|=r}} \prod_{i \in A} x_i.$$

Fix one monomial of type $(2, 1^{a+b})$. It uses $a+b+1$ distinct variables: one variable is squared and the other $a+b$ variables occur to the first power.

To obtain this monomial from $m_{1^{a+1}}m_{1^{b+1}}$, the squared variable must be selected in both factors. Of the remaining $a+b$ variables, choose a to join it in the first factor; the remaining b must be selected in the second factor. There are

$$\binom{a+b}{a} = \binom{a+b}{b}$$

ways to do this. By symmetry every monomial in $m_{2,1^{a+b}}$ has the same coefficient. Therefore the requested coefficient is

$$\boxed{\binom{a+b}{a}}.$$

9. **Solution.**

- (a) In forming the product, we choose either 1 or x_i from each factor. A term obtained by choosing r distinct variables is a monomial of type (1^r) . Hence

$$\prod_{i=1}^{\infty} (1 + x_i) = \sum_{r=0}^{\infty} m_{1^r} = 1 + m_1 + m_{1,1} + m_{1,1,1} + \dots$$

- (b) The same argument applies, but choosing $-x_i$ from r factors contributes the sign $(-1)^r$. Therefore,

$$\prod_{i=1}^{\infty} (1 - x_i) = \sum_{r=0}^{\infty} (-1)^r m_{1^r} = 1 - m_1 + m_{1,1} - m_{1,1,1} + \dots$$

10. **Solution.** For each variable, use the geometric series

$$1 + x_i + x_i^2 + \dots = \frac{1}{1 - x_i}.$$

When the products are multiplied out, every monomial $x_1^{a_1} x_2^{a_2} \dots$ of finite total degree occurs exactly once. Grouping monomials whose positive exponents form the same partition λ gives

$$\prod_{i=1}^{\infty} \frac{1}{1 - x_i} = 1 + \sum_{n=1}^{\infty} \sum_{\lambda \vdash n} m_{\lambda}.$$

The initial 1 corresponds to choosing 1 from every factor, or equivalently to the empty partition. Since the worksheet specifies partitions of positive integers, the requested sum is therefore

$$\sum_{n=1}^{\infty} \sum_{\lambda \vdash n} m_{\lambda} = \prod_{i=1}^{\infty} \frac{1}{1 - x_i} - 1.$$

If the empty partition is included and its monomial symmetric function is defined to be 1, then the answer is simply $\prod_{i=1}^{\infty} (1 - x_i)^{-1}$.

7 Solutions to Worksheet 2

1. **Solution.** For the change of basis from e to m , the number $M_{\lambda,\mu}(e,m)$ counts fillings of the rows of shape λ with content μ such that every row is *strictly increasing*. There is no condition down the columns. Rows in different positions are regarded as different, even when they have the same length.

Here the shape $(2^2, 1^4) = (2, 2, 1, 1, 1, 1)$ has two rows of length 2 and four one-cell rows. Call the long rows A and B . The content $(3^2, 1^2) = (3, 3, 1, 1)$ means that the complete filling must contain three 1's, three 2's, one 3, and one 4. Because a long row is strictly increasing, it cannot contain two equal entries.

We count according to where the exceptional entries 3 and 4 occur.

- (a) Suppose neither 3 nor 4 is in a long row. Then both long rows must be $[1, 2]$. The four one-cell rows contain 1, 2, 3, 4 in any order, giving

$$4! = 24$$

fillings.

- (b) Suppose exactly one of 3, 4 is in a long row. Choose which exceptional entry is used (2 choices), choose whether it is in A or B (2 choices), and choose whether its companion is 1 or 2 (2 choices). The other long row is then forced to be $[1, 2]$. The four one-cell rows contain two equal entries and two other entries, so they can be filled in

$$\frac{4!}{2!} = 12$$

ways. This case contributes

$$2 \cdot 2 \cdot 2 \cdot 12 = 96.$$

- (c) Suppose both 3 and 4 occur in the long rows. There are three possibilities.

- i. If 3 and 4 are together, choose their long row in 2 ways. The other long row must be $[1, 2]$, and the one-cell rows contain 1, 1, 2, 2. This gives

$$2 \cdot \frac{4!}{2!2!} = 12.$$

- ii. If 3 and 4 are in different long rows and have the same companion, choose that companion in 2 ways and choose which long row contains 3 in 2 ways. The one-cell rows contain three copies of one of 1, 2 and one copy of the other, so this gives

$$2 \cdot 2 \cdot \frac{4!}{3!} = 16.$$

- iii. If 3 and 4 are in different long rows and have different companions, choose which long row contains 3 in 2 ways and choose which exceptional entry is paired with 1 in 2 ways. The one-cell rows contain 1, 1, 2, 2, so this gives

$$2 \cdot 2 \cdot \frac{4!}{2!2!} = 24.$$

Thus this case contributes $12 + 16 + 24 = 52$.

Adding the three cases, we obtain

$$M_{(2^2, 1^4), (3^2, 1^2)}(e, m) = 24 + 96 + 52 = 172.$$

2. **Solution.** We use row-strict tableaux of shape λ and content 1^n . The content $1^n = (1, 1, \dots, 1)$ means that the entries $1, 2, \dots, n$ each appear exactly once.

First choose which λ_1 entries go in the first row, then which λ_2 of the remaining entries go in the second row, and so on. Once the entries of a row have been chosen, there is only one legal way to place them: in increasing order. Therefore

$$M_{\lambda, 1^n}(e, m) = \binom{n}{\lambda_1, \dots, \lambda_\ell} = \frac{n!}{\lambda_1! \cdots \lambda_\ell!}.$$

3. **Solution.** The content (n) means that every one of the n cells must contain 1. A strictly increasing row cannot contain two 1's. Therefore every row must have length 1, which is exactly the shape $\lambda = 1^n$. For this shape there is one filling: put 1 in every cell. Hence

$$M_{\lambda, n}(e, m) = \begin{cases} 1, & \lambda = 1^n, \\ 0, & \text{otherwise.} \end{cases}$$

4. **Solution.** Assume $n \geq 2$. The content $(n-1, 1)$ means that the tableau has $n-1$ entries equal to 1 and one entry equal to 2.

Since rows are strictly increasing, no row can contain two 1's. Thus every row has length 1, except possibly one row of length 2. Consequently, the only possible shapes are 1^n and $(2, 1^{n-2})$.

For shape 1^n , choose which of the n one-cell rows contains the unique 2; all other rows contain 1. This gives n fillings. For shape $(2, 1^{n-2})$, the long row is forced to be $[1, 2]$, and every one-cell row is forced to contain 1, giving one filling. Therefore

$$M_{\lambda, (n-1, 1)}(e, m) = \begin{cases} n, & \lambda = 1^n, \\ 1, & \lambda = (2, 1^{n-2}), \\ 0, & \text{otherwise.} \end{cases}$$

5. **Solution.** The statement is false. Take

$$\lambda = (4, 1, 1), \quad \mu = (3, 3).$$

Both are partitions of 6, and $\lambda >_{\text{lex}} \mu$ because $4 > 3$. Their conjugates are

$$\lambda' = (3, 1, 1, 1), \quad \mu' = (2, 2, 2).$$

Now $\mu' <_{\text{lex}} \lambda'$, rather than $\mu' >_{\text{lex}} \lambda'$. This is a counterexample.

6. **Solution.** Let

$$E(t) = \prod_{i \geq 1} (1 + x_i t) = \sum_{r \geq 0} e_r t^r$$

and let ω be a primitive cube root of unity, so $1 + \omega + \omega^2 = 0$. Since

$$1 + x + x^2 = (1 - \omega x)(1 - \omega^2 x),$$

we obtain

$$\prod_{i \geq 1} (1 + x_i + x_i^2) = E(-\omega)E(-\omega^2) = \left(\sum_{r \geq 0} (-\omega)^r e_r \right) \left(\sum_{s \geq 0} (-\omega^2)^s e_s \right).$$

This expression has integer coefficients because the two factors are complex conjugates. For example, its first homogeneous pieces are

$$1 + e_1 + (e_1^2 - e_2) + (e_1 e_2 - 2e_3) + \cdots$$

7. **Solution.**

- (a) Only the first five rows and first three columns can be nonzero. The row sums are $(3, 1, 1, 1, 1)$ and the column sums are $(4, 2, 1)$. The row of sum 3 must be $(1, 1, 1)$. After removing it, the four remaining rows each contain one 1, and the remaining column sums are $(3, 1, 0)$. Choose which three of the four rows place their 1 in the first column; the remaining row uses the second column. Thus

$$M_{(4,2,1),(3,1^4)}(e, m) = \binom{4}{3} = 4.$$

- (b) Transposition is a bijection. A matrix with row sums μ_i and column sums λ_j is sent by transposition to a matrix with row sums λ_j and column sums μ_i . Therefore

$$M_{\lambda, \mu}(e, m) = M_{\mu, \lambda}(e, m).$$

- (c) In the tableaux interpretation, row j of the Young diagram of λ is filled with distinct positive integers, written in increasing order, and integer i occurs μ_i times. Define $a_{ij} = 1$ when integer i occurs in tableau row j , and define $a_{ij} = 0$ otherwise. Distinctness within a tableau row makes this a $\{0, 1\}$ -matrix. Its row sum is μ_i and its column sum is λ_j .

Conversely, from such a matrix, fill tableau row j with the indices i for which $a_{ij} = 1$, in increasing order. These constructions are inverse to one another, so they give the required bijection.

8. **Solution.** For the change of basis from h to m , the number $M_{\lambda, \mu}(h, m)$ counts fillings of the rows of shape λ with content μ such that every row is *weakly increasing*. Again, there is no condition down the columns, and equal-length rows in different positions are distinguished.

We have two long rows A, B of length 2 and four one-cell rows. We must place three 1's, three 2's, one 3, and one 4. The main difference from Exercise 1 is that repeated entries are now allowed in a long row: for example, $[1, 1]$ and $[2, 2]$ are legal.

We count according to how many of 3, 4 occur in the long rows.

- (a) Suppose neither 3 nor 4 is in a long row. Each of A, B is one of

$$[1, 1], \quad [1, 2], \quad [2, 2].$$

The pair $([1, 1], [1, 1])$ is impossible because it uses four 1's, and $([2, 2], [2, 2])$ is impossible for the same reason with 2's. The legal possibilities contribute as follows:

types of A, B	orders	entries in one-cell rows	count
$\{[1, 1], [1, 2]\}$	2	2, 2, 3, 4	$2 \cdot 4! / 2! = 24$
$\{[1, 1], [2, 2]\}$	2	1, 2, 3, 4	$2 \cdot 4! = 48$
$\{[1, 2], [1, 2]\}$	1	1, 2, 3, 4	$4! = 24$
$\{[1, 2], [2, 2]\}$	2	1, 1, 3, 4	$2 \cdot 4! / 2! = 24$

Thus this case contributes 120.

- (b) Suppose exactly one of 3, 4 is in a long row. Choose that exceptional entry in 2 ways and choose its long row in 2 ways. Its companion is either 1 or 2.

If its companion is 1, the other long row may be $[1, 1]$, $[1, 2]$, or $[2, 2]$. The corresponding one-cell rows can be filled in

$$\frac{4!}{3!} = 4, \text{quad} \frac{4!}{2!} = 12, \text{quad} \frac{4!}{2!} = 12$$

ways, respectively. This gives $4 + 12 + 12 = 28$ fillings. If the companion is 2, the same argument with 1 and 2 interchanged gives another 28. Therefore this case contributes

$$2 \cdot 2 \cdot (28 + 28) = 224.$$

(c) Suppose both 3 and 4 are in the long rows.

- i. They may be together in one long row, which can be chosen in 2 ways. The other long row may be $[1, 1]$, $[1, 2]$, or $[2, 2]$. The one-cell rows can then be filled in 4, 6, or 4 ways, respectively. This gives

$$2(4 + 6 + 4) = 28.$$

- ii. They may be in different long rows. There are 2 ways to decide which long row contains 3. If both companions are 1, the one-cell rows can be filled in 4 ways; if both are 2, there are 4 ways; and if the companions are different, there are 2 choices for their assignment and 6 fillings of the one-cell rows. This gives

$$2(4 + 4 + 2 \cdot 6) = 40.$$

Thus this case contributes $28 + 40 = 68$.

Adding all three cases gives

$$M_{(2^2, 1^4), (3^2, 1^2)}(h, m) = 120 + 224 + 68 = 412.$$

9. **Solution.** We count weakly increasing row fillings of shape λ and content 1^n . Since the entries $1, 2, \dots, n$ are all distinct, choose which λ_1 entries go in the first row, which λ_2 of the remaining entries go in the second row, and so forth. After the entries of a row have been chosen, their weakly increasing order is forced; because the entries are distinct, it is actually strictly increasing. Thus

$$M_{\lambda, 1^n}(h, m) = \frac{n!}{\lambda_1! \cdots \lambda_\ell!}.$$

10. **Solution.** The content (n) means that every cell contains 1. This is allowed because a row only has to be weakly increasing, so repeated entries are permitted. Every row is therefore forced, and there is exactly one tableau of every shape $\lambda \vdash n$. Thus

$$M_{\lambda, n}(h, m) = 1 \quad \text{for every } \lambda \vdash n.$$

11. **Solution.** The content $(n-1, 1)$ consists of $n-1$ copies of 1 and one copy of 2. Choose the row that will contain the unique 2. In that row, weak increase forces the 2 to be the final entry, with all earlier entries equal to 1. Every other row consists entirely of 1's.

Thus each row gives exactly one tableau, depending on whether that row is chosen to contain the 2. Since λ has $\ell(\lambda)$ rows, including rows of equal length in different positions, we obtain

$$M_{\lambda, (n-1, 1)}(h, m) = \ell(\lambda),$$

where $\ell(\lambda)$ is the number of parts of λ .

12. **Solution.** Write

$$p_\lambda = p_{\lambda_1} \cdots p_{\lambda_\ell}, \quad p_{\lambda_j} = \sum_{i \geq 1} x_i^{\lambda_j}.$$

For each row j , choosing the term $x_i^{\lambda_j}$ from p_{λ_j} means placing the entry i in all λ_j cells of that row. Therefore the relevant tableaux are exactly the fillings in which *every row is constant*. There is no condition relating one row to another.

After making one choice for every row, the exponent of x_i is the total number of cells filled with i . Hence the resulting monomial has exponent sequence μ exactly when the filling has content μ . We have therefore proved the combinatorial interpretation

$$M_{\lambda, \mu}(p, m) = \#\{\text{constant-row fillings of shape } \lambda \text{ and content } \mu\}.$$

13. **Solution.** We count constant-row tableaux of shape $(2, 2, 1, 1, 1, 1)$ and content $(3, 3, 1, 1)$. Thus we need three 1's, three 2's, one 3, and one 4.

The two long rows cannot be filled with 3 or 4, because a constant row of length 2 would use two copies of that entry, but the content provides only one copy of each. They also cannot both be filled with 1, since that would require four 1's, and they cannot both be filled with 2. Therefore one long row must be $[1, 1]$ and the other must be $[2, 2]$.

There are 2 ways to decide which long row gets the 1's. After that choice, the four one-cell rows must contain one each of 1, 2, 3, 4, and these four entries can be assigned to the four rows in $4!$ ways. Hence

$$M_{(2^2, 1^4), (3^2, 1^2)}(p, m) = 2 \cdot 4! = 48.$$

14. **Solution.** The content 1^n uses each of the entries $1, 2, \dots, n$ exactly once. A constant row of length greater than 1 would repeat its entry, so such a row is impossible. Consequently every row must have length 1, which means $\lambda = 1^n$.

For this shape, assign the n distinct entries to the n one-cell rows. The first row has n choices, the second has $n - 1$, and so on, for a total of $n!$ tableaux. Therefore

$$M_{\lambda, 1^n}(p, m) = \begin{cases} n!, & \lambda = 1^n, \\ 0, & \text{otherwise.} \end{cases}$$

15. **Solution.** The content (n) means that all n cells contain 1. Every row is then constant automatically, and there are no choices to make. Thus there is exactly one such tableau for every shape $\lambda \vdash n$, and

$$M_{\lambda, n}(p, m) = 1 \quad \text{for every } \lambda \vdash n.$$

16. **Solution.** The content $(n-1, 1)$ consists of $n-1$ copies of 1 and one copy of 2. Because rows must be constant, the row containing 2 cannot have length greater than 1: otherwise it would contain at least two 2's. Thus the unique 2 must occupy a one-cell row. After choosing that row, every other row is forced to contain only 1's.

There is one choice for each part of λ equal to 1. If $m_1(\lambda)$ denotes the number of such parts, then

$$\boxed{M_{\lambda, (n-1, 1)}(p, m) = m_1(\lambda),}$$

where $m_1(\lambda)$ denotes the multiplicity of the part 1 in λ .

8 Solutions to Worksheet 3

1. **Solution.** The semistandard tableaux of shape $(2, 2)$ give

$$s_{2,2} = m_{2,2} + m_{2,1,1} + 2m_{1,1,1,1}.$$

For example, the coefficient 2 is the number of standard Young tableaux of shape $(2, 2)$.

The Jacobi-Trudi identity gives

$$s_{2,2} = h_2^2 - h_3 h_1 = h_{2,2} - h_{3,1}.$$

Since $(2, 2)$ is self-conjugate and the involution ω exchanges h_r with e_r , we also have

$$s_{2,2} = e_2^2 - e_3 e_1 = e_{2,2} - e_{3,1}.$$

Finally, using the character values for the representation indexed by $(2, 2)$, or converting the preceding formulas with Newton's identities, gives

$$s_{2,2} = \frac{1}{12}p_{1,1,1,1} + \frac{1}{4}p_{2,2} - \frac{1}{3}p_{3,1}.$$

2. **Solution.**

- (a) Suppose a semistandard tableau of shape λ and content μ exists. Because columns strictly increase, an entry in row r is at least r . Therefore all entries among $1, 2, \dots, r$ must lie in the first r rows. It follows that

$$\mu_1 + \dots + \mu_r \leq \lambda_1 + \dots + \lambda_r \quad \text{for every } r.$$

If $\mu >_{\text{lex}} \lambda$, let r be the first index at which they differ. Then $\mu_i = \lambda_i$ for $i < r$ and $\mu_r > \lambda_r$, contradicting the displayed inequality. Hence

$$\mu >_{\text{lex}} \lambda \implies K_{\lambda, \mu} = 0.$$

- (b) In a tableau of shape and content λ , all λ_1 copies of 1 must lie in the first row, so that row consists entirely of 1's. After removing it, all λ_2 copies of 2 must fill the second row. Continuing in this way shows that row i must consist entirely of i 's. This tableau is semistandard and is the only possibility. Thus

$$K_{\lambda, \lambda} = 1.$$

3. **Solution.** The monomial expansion of a Schur function is

$$s_\lambda = \sum_{\mu \vdash k} K_{\lambda, \mu} m_\mu.$$

Order the partitions of k lexicographically. Exercise 2 says that this change-of-basis matrix is triangular, and its diagonal entries are all $K_{\lambda,\lambda} = 1$. Its determinant is therefore 1, so it is invertible. Since the monomial functions $\{m_\mu : \mu \vdash k\}$ form a basis of Λ^k , the same is true of

$$\boxed{\{s_\lambda : \lambda \vdash k\}}.$$

4. **Solution.** Only the entries 1 and 2 are available. Since columns strictly increase, a tableau can have at most two rows. Its shape must be

$$(2k - r, r) \quad \text{for some } 0 \leq r \leq k.$$

In the r columns of height two, the top entry must be 1 and the bottom entry must be 2. The remaining cells are all in the first row; weak increase forces the remaining $k - r$ copies of 1 to come before the remaining $k - r$ copies of 2. Thus there is exactly one tableau for each value of r . The answer is

$$\boxed{k + 1}.$$

5. **Solution.** Let

$$E(t) = \sum_{n \geq 0} e_n t^n, \quad H(t) = \sum_{n \geq 0} h_n t^n.$$

The given identity is exactly the statement

$$E(-t)H(t) = 1.$$

Suppose ω is defined by $\omega(e_n) = h_n$ and extended as an algebra homomorphism. Apply ω to the displayed identity:

$$H(-t)\omega(H(t)) = 1.$$

Replacing t by $-t$ in the original identity also gives $E(t)H(-t) = 1$. An inverse is unique, so

$$\omega(H(t)) = E(t).$$

Comparing coefficients yields $\omega(h_n) = e_n$. Thus ω^2 fixes every e_n (and every h_n). Since the e_n generate the ring of symmetric functions,

$$\boxed{\omega^2 = \text{id}}.$$

6. **Solution.** Since $h_r = s_r$, apply the Pieri rule to $s_3 h_2$. Adding a horizontal strip of two boxes to the row (3) produces exactly (5), (4, 1), and (3, 2). Hence

$$\boxed{h_{3,2} = s_5 + s_{4,1} + s_{3,2}}.$$

7. **Solution.** Apply ω to the answer in Exercise 6. The involution exchanges h and e and sends s_λ to $s_{\lambda'}$. The conjugates of (5) , $(4, 1)$, and $(3, 2)$ are respectively (1^5) , $(2, 1^3)$, and $(2, 2, 1)$. Therefore

$$e_{3,2} = s_{1^5} + s_{2,1^3} + s_{2,2,1}.$$

8. **Solution.** Newton's identities give

$$p_3 = 3h_3 - 3h_2h_1 + h_1^3.$$

By the Pieri rule,

$$h_3 = s_3, \quad h_2h_1 = s_3 + s_{2,1}, \quad h_1^3 = s_3 + 2s_{2,1} + s_{1,1,1}.$$

Substitution gives

$$p_3 = s_3 - s_{2,1} + s_{1,1,1}.$$

9. **Solution.** The statement is false. Let $\lambda = (2)$ and $\mu = (1, 1)$. There is one semistandard tableau of shape (2) and content $(1, 1)$, namely a row containing 1, 2. Thus $K_{(2),(1,1)} = 1$. A column of shape $(1, 1)$ cannot be filled with two copies of 1 because columns must strictly increase, so $K_{(1,1),(2)} = 0$. Hence the two Kostka numbers need not be equal.
10. **Solution.** The partitions of 3 give

$$\begin{aligned} h_3 &= s_3, \\ h_{2,1} &= s_3 + s_{2,1}, \\ h_{1,1,1} &= s_3 + 2s_{2,1} + s_{1,1,1}. \end{aligned}$$

The pattern is Young's rule:

$$h_\lambda = \sum_{\mu \vdash 3} K_{\mu,\lambda} s_\mu.$$

In general, the coefficient of s_μ in h_λ is the number of semistandard tableaux of shape μ and content λ .

11. **Solution.** For $a \geq 1$ and $b \geq 0$, the Jacobi–Trudi determinant for the hook $(a, 1^b)$ is

$$s_{a,1^b} = \begin{vmatrix} h_a & h_{a+1} & h_{a+2} & \cdots & h_{a+b} \\ 1 & h_1 & h_2 & \cdots & h_b \\ 0 & 1 & h_1 & \cdots & h_{b-1} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & h_1 \end{vmatrix}.$$

Expand along the first row. The minor corresponding to h_{a+i} is the Jacobi–Trudi determinant for e_{b-i} ; the cofactor sign is $(-1)^i$. Therefore

$$s_{a,1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}.$$

This also proves the formula inductively: the case $b = 0$ is $s_a = h_a$, and increasing b adds one more cofactor to the same Jacobi–Trudi expansion.

9 Solutions to Worksheet 4

1. **Solution.** Recall that

$$X_G = \sum_{\kappa} \prod_{v \in V(G)} x_{\kappa(v)},$$

where the sum is over proper colorings by positive integers. Let σ be any permutation of the positive integers. Replacing every color i by $\sigma(i)$ is a bijection from proper colorings to proper colorings. It permutes the variables in every monomial but leaves the whole sum unchanged. Thus X_G is invariant under every permutation of the variables, so it is symmetric.

2. **Solution.** Let e be any edge of C_4 . Deleting e produces the path P_4 , which is a tree on four vertices, while contracting e produces the triangle C_3 . Therefore

$$\chi_{C_4}(n) = n(n-1)^3 - n(n-1)(n-2).$$

Simplifying,

$$\chi_{C_4}(n) = n(n-1)(n^2 - 3n + 3) = n^4 - 4n^3 + 6n^2 - 3n.$$

3. **Solution.** We use Stanley's power-sum formula

$$X_G = \sum_{A \subseteq E(G)} (-1)^{|A|} p_{\lambda(A)},$$

where $\lambda(A)$ is the partition of the component sizes of the spanning subgraph $(V(G), A)$.

- (a) Deleting the chosen edge gives P_4 . Among its three edges, there are three one-edge subsets, two adjacent two-edge subsets, one disjoint two-edge subset, and one three-edge subset. Hence

$$X_{P_4} = p_{1,1,1,1} - 3p_{2,1,1} + 2p_{3,1} + p_{2,2} - p_4.$$

- (b) Contracting the edge gives $C_3 = K_3$. The subset formula gives

$$X_{C_3} = p_{1,1,1} - 3p_{2,1} + 2p_3.$$

- (c) The two terms in a deletion-contraction formula do not even have the same degree: X_{P_4} has degree 4, while X_{C_3} has degree 3. More explicitly,

$$X_{C_4} = p_{1,1,1,1} - 4p_{2,1,1} + 4p_{3,1} + 2p_{2,2} - 3p_4,$$

which is not $X_{P_4} - X_{C_3}$. Thus the ordinary deletion-contraction relation does not hold for the chromatic symmetric function.

4. **Solution.**

(a) By the binomial theorem,

$$\begin{aligned} n(n-1)^{k-1} &= n \sum_{i=0}^{k-1} \binom{k-1}{i} n^i (-1)^{k-1-i} \\ &= \sum_{j=1}^k (-1)^{k-j} \binom{k-1}{j-1} n^j. \end{aligned}$$

(b) Taking $j = k - 1$, the coefficient is

$$-\binom{k-1}{k-2} = -(k-1).$$

(c) There is a missing minus sign in the worksheet statement. For a simple graph,

$$[n^{k-1}] \chi_G(n) = -|E(G)|.$$

We prove this by induction on the number of edges. For the graph with no edges, $\chi_G(n) = n^k$, and both sides are 0. Choose an edge e . By induction, the coefficient of n^{k-1} in $\chi_{G-e}(n)$ is $-(|E(G)| - 1)$. The contracted graph has $k - 1$ vertices, so its chromatic polynomial is monic of degree $k - 1$. Consequently, deletion-contraction gives

$$[n^{k-1}] \chi_G(n) = -(|E(G)| - 1) - 1 = -|E(G)|.$$

(d) If $\chi_G(n) = n(n-1)^{k-1}$, part (b) says that the coefficient of n^{k-1} is $-(k-1)$. By part (c), G has $k-1$ edges. Since G is connected, a connected graph on k vertices with $k-1$ edges is a tree.

5. **Solution.** Call the graphs G_1 and G_2 in their displayed order. In either graph there are four independent pairs. Thus a coloring of type $(2, 1, 1, 1)$ is obtained by choosing one of those pairs and then assigning the three singleton colors in $3!$ ways, giving coefficient $4 \cdot 3! = 24$.

There are two ways in either graph to choose two disjoint independent pairs. Since the two pair-colors can be interchanged, the coefficient of $m_{2,2,1}$ is $2 \cdot 2! = 4$. No independent set has size 3, and a coloring with all colors distinct contributes $5! = 120$. Therefore the two graphs have the same answer:

$$X_{G_1} = X_{G_2} = 4m_{2,2,1} + 24m_{2,1,1,1} + 120m_{1,1,1,1,1}.$$

6. **Solution.** In the power expansion, the coefficient of p_n is

$$\sum_{\substack{A \subseteq E(G) \\ (V(G), A) \text{ connected}}} (-1)^{|A|}.$$

It is zero when G is disconnected, because no edge subset can connect all the vertices. When G is connected, the signed sum is nonzero; up to the sign $(-1)^{n-1}$, it counts acyclic orientations having a unique sink at any fixed vertex. At least one such orientation is obtained by ordering vertices by their distance from the chosen sink and directing edges toward smaller distance. Consequently,

$$\boxed{G \text{ is connected if and only if } [p_n]X_G \neq 0.}$$

More generally, the smallest number of parts among partitions indexing nonzero power-sum terms is the number of connected components.

7. **Solution.** The term $p_{2,1^{n-2}}$ in the subset expansion can arise only from a subset containing exactly one edge. Every edge contributes $-p_{2,1^{n-2}}$. Hence

$$\boxed{[p_{2,1^{n-2}}]X_G = -|E(G)|.}$$

If $X_G = X_H$, these coefficients agree, so G and H have the same number of edges.

8. **Solution.** In a proper coloring of K_n , all vertices receive distinct colors. For each set of n colors there are $n!$ assignments to the vertices, so

$$\boxed{X_{K_n} = n!e_n.}$$

In particular,

$$X_{K_1} = e_1, \quad X_{K_2} = 2e_2, \quad X_{K_3} = 6e_3, \quad X_{K_4} = 24e_4, \quad X_{K_5} = 120e_5.$$

9. **Solution.** Let $e = uv$, where u is the leaf and v is its neighbor.

- (a) Deleting e leaves u isolated. Thus $G \setminus e = (G - u) \sqcup K_1$.
- (b) Dot-contraction contracts u and v and then adds an isolated vertex. Contracting a leaf edge simply removes the leaf, so the result is again $(G - u) \sqcup K_1$.
- (c) The graphs in (a) and (b) are isomorphic.
- (d) Leaf-contraction contracts u and v and then attaches a new leaf to the merged vertex. This recreates a graph isomorphic to G .
- (e) The deletion-near-contraction relation is

$$X_G = X_{G \setminus e} - X_{\text{dot}(G,e)} + X_{\text{leaf}(G,e)}.$$

For a leaf edge, the first two terms cancel and the last term is X_G , so the relation reduces to the tautology $X_G = X_G$. It makes no progress. This is why the recursive algorithm uses internal edges.

10. **Solution.** Write $\text{st}_r = X_{St_r}$ and $\text{st}_\lambda = \prod_i \text{st}_{\lambda_i}$. There are three trees on five vertices: St_5 , the tree T_5 with degree sequence $(3, 2, 1, 1, 1)$, and the path P_5 .

For T_5 , apply DNC to the unique internal edge adjacent to the degree-3 vertex. The deletion, dot-contraction, and leaf-contraction terms are respectively star forests of types $(3, 2)$, $(4, 1)$, and (5) . Thus

$$X_{T_5} = \text{st}_5 - \text{st}_{4,1} + \text{st}_{3,2}.$$

Similarly, applying DNC to the middle edge of P_4 gives

$$X_{P_4} = \text{st}_{2,2} - \text{st}_{3,1} + \text{st}_4.$$

Applying DNC to an internal edge of P_5 then gives

$$\begin{aligned} X_{P_5} &= \text{st}_{3,2} - X_{P_4}\text{st}_1 + X_{T_5} \\ &= \text{st}_5 - 2\text{st}_{4,1} + 2\text{st}_{3,2} + \text{st}_{3,1,1} - \text{st}_{2,2,1}. \end{aligned}$$

Finally, by definition,

$$X_{St_5} = \text{st}_5.$$

11. **Solution.** Call the displayed graphs H_1 and H_2 . The subset formula gives the same power-sum expansion for both:

$$\begin{aligned} X_{H_i} &= p_1^6 - 6p_{2,1^4} + 6p_{2,2,1,1} - p_{2,2,2} + 8p_{3,1,1,1} \\ &\quad - 6p_{3,2,1} - 9p_{4,1,1} + 3p_{4,2} + 6p_{5,1} - 2p_6. \end{aligned}$$

To convert to the star basis, one may use

$$p_n = \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} \text{st}_{r+1, 1^{n-1-r}}$$

and multiply this identity for the parts of each power-sum index. After collecting terms,

$$\begin{aligned} X_{H_1} = X_{H_2} &= 2\text{st}_6 - 4\text{st}_{5,1} + 3\text{st}_{4,2} + 2\text{st}_{4,1,1} \\ &\quad - 3\text{st}_{3,2,1} + \text{st}_{2,2,2}. \end{aligned}$$

Thus these two nonisomorphic graphs have the same chromatic symmetric function.

10 Solutions to the SageMath Worksheet

The computations below were checked with SageMath 10.8. We begin every session by creating the four bases that occur in the worksheet.

```
Sym = SymmetricFunctions(QQ)
m = Sym.monomial()
e = Sym.elementary()
p = Sym.powersum()
s = Sym.schur()
```

Calling, for example, `e(X)` converts the symmetric function `X` to the elementary basis.

1. **Solution.** The following code computes all the requested matrices.

```
for n in range(1, 7):
    order = list(Partitions(n))
    A = m.transition_matrix(s, n)
    upper = all(A[i,j] == 0
                for i in range(A.nrows())
                for j in range(i))
    print("n =", n, "order =", order)
    print(A)
    print("upper triangular:", upper)
    print("determinant:", A.det())
```

The method `m.transition_matrix(s,n)` writes the degree- n monomial basis in the Schur basis. Sage orders partitions in decreasing lexicographic order. The matrices are

n=1: [1]

n=2: [1 -1]	n=3: [1 -1 1]
[0 1]	[0 1 -2]
	[0 0 1]

n=4:
[1 -1 0 1 -1]
[0 1 -1 -1 2]
[0 0 1 -1 1]
[0 0 0 1 -3]
[0 0 0 0 1]

n=5:
[1 -1 0 1 0 -1 1]
[0 1 -1 -1 1 1 -2]

```

[0  0  1 -1 -1  2 -2]
[0  0  0  1 -1 -1  3]
[0  0  0  0  1 -2  3]
[0  0  0  0  0  1 -4]
[0  0  0  0  0  0  1]

```

n=6:

```

[1 -1  0  1  0  0 -1  0  0  1 -1]
[0  1 -1 -1  0  1  1  0 -1 -1  2]
[0  0  1 -1 -1  0  1 -1  1 -2  2]
[0  0  0  1  0 -1 -1  1  1  1 -3]
[0  0  0  0  1 -1  1  1  0 -1  1]
[0  0  0  0  0  1 -2 -2  0  4 -6]
[0  0  0  0  0  0  1  0 -1 -1  4]
[0  0  0  0  0  0  0  1 -1  1 -1]
[0  0  0  0  0  0  0  0  1 -3  6]
[0  0  0  0  0  0  0  0  0  1 -5]
[0  0  0  0  0  0  0  0  0  0  1]

```

Every matrix is upper triangular with diagonal entries 1, so every determinant is 1. This is the computational form of the unitriangularity of the Kostka matrix.

2. **Solution.** Here are examples with fewer than four parts, exactly four parts, and more than four parts.

```

examples = [[2], [2,1], [1,1,1,1], [1,1,1,1,1]]
for la in examples:
    print(la, s[la].expand(4))
    print("monomial form:", m(s[la]))

```

Sage uses variables x_0, x_1, x_2, x_3 . The mathematical results are

$$s_2 = m_2 + m_{1,1}, \quad s_{2,1} = m_{2,1} + 2m_{1,1,1}, \quad s_{1^4} = m_{1^4}.$$

The abstract function $s_{1^5} = m_{1^5}$ is nonzero. In contrast, the command

```
s[1,1,1,1,1].expand(4)
```

returns 0, because a Schur polynomial whose shape has more than four rows vanishes in four variables.

Thus `expand(4)` realizes an abstract symmetric function as a symmetric polynomial in four variables. The monomial coefficients are Kostka numbers: they are nonnegative integers counting semistandard tableaux.

3. **Solution.** The following loop performs every requested calculation.

```

families = [
    ("paths",    range(1,8), graphs.PathGraph),
    ("cycles",   range(3,8), graphs.CycleGraph),
    ("complete", range(1,8), graphs.CompleteGraph)
]
bases = [("m",m), ("e",e), ("p",p), ("s",s)]

for name, sizes, constructor in families:
    print("\n", name)
    for n in sizes:
        G = constructor(n)
        X = G.chromatic_symmetric_function(QQ)
        print("n =", n)
        for basis_name, B in bases:
            print(basis_name, B(X))

chromatic_symmetric_function() first returns the power-sum expansion, using

```

$$X_G = \sum_{A \subseteq E(G)} (-1)^{|A|} p_{\lambda(A)}.$$

The numbers of nonzero terms in the four bases are:

family and sizes	basis	numbers of terms
$P_n, 1 \leq n \leq 7$	m, s	(1, 1, 2, 3, 5, 7, 11)
	e	(1, 1, 2, 3, 4, 6, 8)
	p	(1, 2, 3, 5, 7, 11, 15)
$C_n, 3 \leq n \leq 7$	m, s	(1, 3, 3, 7, 8)
	e	(1, 2, 2, 4, 4)
	p	(3, 5, 7, 11, 15)
$K_n, 1 \leq n \leq 7$	m, e, s	(1, 1, 1, 1, 1, 1, 1)
	p	(1, 2, 3, 5, 7, 11, 15)

For paths and cycles in this range, the elementary expansion has the fewest terms. For complete graphs,

$$X_{K_n} = n!e_n,$$

so the elementary basis is especially natural. All three families have non-negative coefficients in the monomial, elementary, and Schur bases for these values of n . Negative power-sum coefficients occur as soon as $n \geq 2$.

4. **Solution.** Sage's `StarGraph(r)` has one center and r leaves, so it has $r + 1$ vertices.

```

for n in range(1, 8):
    St = graphs.StarGraph(n-1)
    X = p(St.chromatic_symmetric_function(QQ))

```

```

formula = sum((-1)^r * binomial(n-1,r)
              * p[Partition([r+1] + [1]*(n-1-r))])
              for r in range(n))
print(n, X, X == formula)

```

Every comparison returns `True`. Thus

$$X_{St_n} = \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} p_{r+1, 1^{n-1-r}}.$$

This also follows directly from the edge-subset formula: choosing r of the $n-1$ star edges creates one component of size $r+1$ and $n-1-r$ isolated vertices.

5. Solution.

```

from collections import defaultdict

for n in [7,8]:
    groups = defaultdict(list)
    trees = list(graphs.trees(n))
    for i,T in enumerate(trees, start=1):
        X = T.chromatic_symmetric_function(QQ)
        groups[X].append(i)
    collisions = [I for I in groups.values() if len(I) > 1]
    print(n, len(trees), collisions)

```

Sage generates one representative from each isomorphism class. It finds 11 trees for $n = 7$ and 23 trees for $n = 8$. In both cases `collisions` is empty. Thus no two trees in either collection have the same chromatic symmetric function.

6. Solution.

The code below computes the elementary expansion and groups trees by the requested coefficient.

```

trees8 = list(graphs.trees(8))
groups = defaultdict(list)

for i,T in enumerate(trees8, start=1):
    XE = e(T.chromatic_symmetric_function(QQ))
    c = XE[Partition([4,2,1,1])]
    groups[c].append(i)
    print(i, XE, "coefficient =", c)

print({c:I for c,I in groups.items() if len(I) > 1})

```

With trees numbered in Sage's iterator order, the repeated groups are

coefficient	tree indices
0	1, 16
3	4, 10
6	2, 7
10	3, 14
12	9, 22
13	8, 18

Therefore the coefficient of $e_{4,2,1,1}$ does not distinguish all trees on eight vertices.

7. **Solution.** This program computes all five requested statistics.

```
for i,T in enumerate(trees8, start=1):
    X = T.chromatic_symmetric_function(QQ)
    XE, XP = e(X), p(X)
    deg = T.degree_sequence()
    leaves = deg.count(1)
    print(i,
          "degree sequence =", deg,
          "diameter =", T.diameter(),
          "leaves =", leaves,
          "e[7,1] =", XE[Partition([7,1])],
          "p[7,1] =", XP[Partition([7,1])])
```

The coefficient pair depends only on the number L of leaves:

L	2	3	4	5	6	7
$[e_{7,1}]X_T$	6	13	20	27	34	41
$[p_{7,1}]X_T$	2	3	4	5	6	7

In fact, for a tree on n vertices,

$$[p_{n-1,1}]X_T = (-1)^{n-2}L, \quad [e_{n-1,1}]X_T = (n-1)L - n.$$

For the power coefficient, a contributing edge subset must isolate one leaf and contain all edges of the remaining $(n-1)$ -vertex tree. Thus these coefficients recover the number of leaves, which is already visible in the degree sequence. They do not determine the diameter: different trees with the same number of leaves can have different diameters.

Mini-Projects

A. **Solution.** The following code produces a reproducible data record for every tree.

```

def basis_statistics(f):
    coeffs = f.coefficients()
    return {
        "terms": len(f.support()),
        "smallest coefficient": min(coeffs),
        "largest coefficient": max(coeffs),
        "all positive": all(c > 0 for c in coeffs)
    }

records = []
for i,T in enumerate(trees8, start=1):
    X = T.chromatic_symmetric_function(QQ)
    row = {"tree": i, "graph6": T.graph6_string()}
    for name,B in [("m",m),("e",e),("p",p),("s",s)]:
        row[name] = basis_statistics(B(X))
    records.append(row)
    print(row)

```

The total numbers of nonzero terms over all 23 trees are

basis	<i>m</i>	<i>e</i>	<i>p</i>	<i>s</i>
total terms	384	377	416	401
average	16.70	16.39	18.09	17.43

There is no basis that is sparsest for every tree. Counting ties, the elementary basis is sparsest for 15 trees, the monomial and power bases for 7 each, and the Schur basis for 3.

All 23 monomial expansions have positive coefficients, as follows directly from the coloring definition. Only one of these trees (the path) has all positive elementary coefficients in this experiment, and 9 have all positive Schur coefficients. None has all positive power-sum coefficients. The coefficients also encode graph data: for example, Exercise 7 shows how to recover the number of leaves.

B. **Solution.** Sage partitions can be sorted directly in lexicographic order.

```

for i,T in enumerate(trees8, start=1):
    X = T.chromatic_symmetric_function(QQ)
    print("tree", i)
    for name,B in [("m",m),("e",e),("p",p),("s",s)]:
        support = sorted(B(X).support())
        print(name, "smallest =", support[0],
              "largest =", support[-1])

```

In the monomial basis, the smallest partition is always 1^8 . The first part of the largest partition is the independence number $\alpha(T)$: a color class is an independent set, and lexicographic maximization first makes the largest

color class as large as possible. Thus the largest monomial index contains genuine information about the tree. In the power basis, connectedness forces a nonzero p_8 term, so the largest index is always (8) and is not useful for distinguishing these trees by itself.

To study the effect of adding a leaf, attach it at every possible vertex and recompute the dictionaries of coefficients:

```
def add_leaf(T, v):
    U = T.copy()
    new_vertex = max(U.vertices(sort=False)) + 1
    U.add_edge(v, new_vertex)
    return U

for i,T in enumerate(trees8, start=1):
    for v in T.vertices(sort=False):
        U = add_leaf(T, v)
        XU = U.chromatic_symmetric_function(QQ)
        print("tree", i, "attachment vertex", v)
        for name,B in [("m",m),("e",e),("p",p),("s",s)]:
            print(name, B(XU).monomial_coefficients())
```

There is no single coefficient change independent of the attachment vertex. Attaching at different vertices can produce nonisomorphic trees and different expansions. The code makes that dependence explicit and repeats the experiment in all four bases.